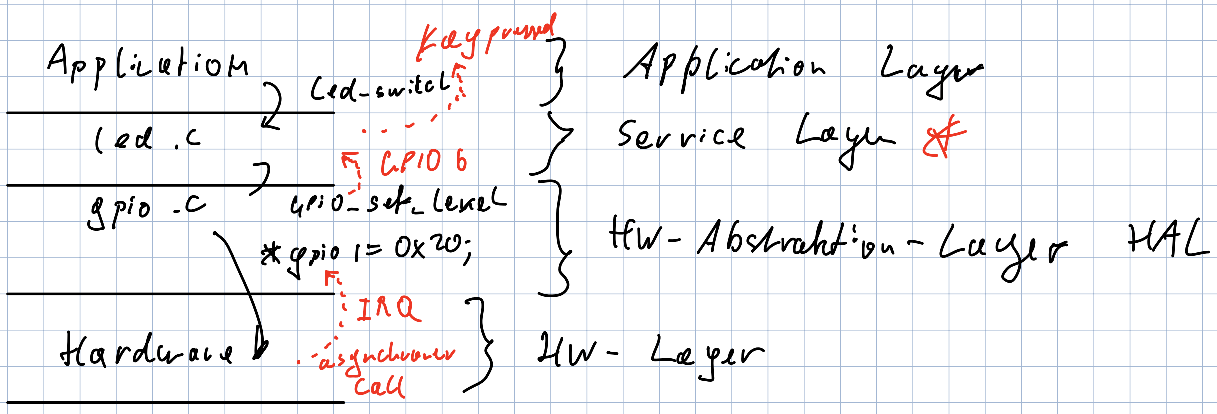
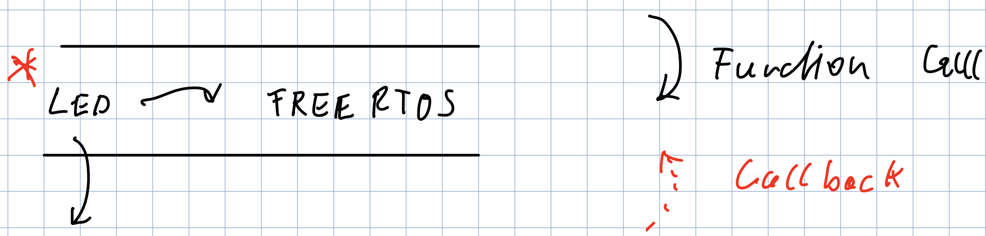
Schichten architektur





in button.c

```
extern void buttonCallback(uint8_t key,
                            keystate ks);
buttonkeyCallback(key, pressed);
```

*

in applikation

```
void buttonkeyCallback(uint8_t key,
                       keystate KS) {

printf("Button pressed");
}
```

=> statischer Callback = "zur Compilezeit"

```c
* __weak__
void buttonKeyCallback (...) {
    //nothing
};
```

<u>Dynamischer Callback</u>

button.h:

```c
typedef void (*ButtonKeyCallback)(uint8_t Key,
                                  Keystate ks);
```

button.c:

```c
static ButtonKeyCallback gBtnKeyCallback = NULL;
```

Rückruf:

```c
if (gBtnKeyCallback != NULL){
    gBtnKeyCallback (Key, Pressed);
};
```

Registrierung

```c
registerKeyCallback (BtnKeyCallback cb ) {
    gBtnKeyCallback = cb;
}

void handleKey (uint8_t key, Keystate ks) {
    printf ("Key pressed");
};

btn_registerKeyCallback(handleKey);
```

Prüfung: Was ist MMIO

Volatile PTR für

Callback statisch dynamisch

Programm das LED blinken lässt

Bib-funktionen

→ Fkt müssen nicht genau passen

1/3 Theorie

2/3 Praxis

Was sind Register

Bedeutung Bus

Interrupt

static
